
GENERAL RELATIVITY

Tutorial problem set 5, 09.10.2020, solutions

■ PROBLEM 13 Causality in GR.

Consider the following three 3-dimensional space-time metrics,

$$(i) \quad ds^2 = -c^2 dt^2 + dr^2 + r^2 d\varphi^2, \quad (13.1)$$

$$(ii) \quad ds^2 = -c^2 dt^2 + \frac{d\rho^2}{1 - \kappa\rho^2} + \rho^2 d\varphi^2, \quad (\kappa > 0 \text{ or } \kappa < 0) \quad (13.2)$$

$$(iii) \quad ds^2 = -c^2 dt^2 + R^2 d\vartheta^2 + R^2 \sin^2(\vartheta) d\varphi^2. \quad (R > 0) \quad (13.3)$$

The ranges of coordinates are $\varphi \in [0, 2\pi)$, $r \in [0, \infty)$, $\vartheta \in [0, \pi)$, $\rho \in [-\frac{1}{\sqrt{\kappa}}, \frac{1}{\sqrt{\kappa}}]$ for $\kappa > 0$ where $\rho = \frac{1}{\sqrt{\kappa}}$ and $\rho = -\frac{1}{\sqrt{\kappa}}$ are identified, $\rho \in [0, \infty]$ for $\kappa < 0$.

- (a) Which manifolds are represented by the metric (A) case (ii) (for $\kappa > 0$) and (B) case (iii)? Can you parametrize a sphere in cylindrical coordinates?
- (b) Derive what are the curves traced by light in these three spaces in given coordinates and sketch the light cones. Note that there is a rotation symmetry in all three metrics with respect to angle φ .
- (c) Sketch the closed achronal surface S and the future and past domains of dependence $D^+(S)$ and $D^-(S)$, if for each of the cases S is defined to be

$$(i) \quad r \in [0, R], \varphi \in [0, 2\pi); \quad (13.4)$$

$$(ii) \quad \kappa > 0: \quad \rho \in [0, \rho_0], \varphi \in [0, 2\pi) \quad (0 < \rho_0 < \frac{1}{\sqrt{\kappa}}) \quad (13.5)$$

$$\kappa < 0: \quad \rho \in [0, \rho_0], \varphi \in [0, 2\pi); \quad (13.6)$$

$$(iii) \quad \vartheta \in [0, \vartheta_0], \varphi \in [0, 2\pi) \quad (0 < \vartheta_0 < \pi). \quad (13.7)$$

- (d) Sketch the Cauchy surface for all the cases.

SOLUTION: (a) Both cases are coordinates on manifolds, $\mathbf{R} \times S^2$, $\mathbf{R}$ parametrizing the time. The case (ii) corresponds to one hemisphere for $1/\sqrt{\kappa} > \rho > 0$ and another hemisphere for $-1/\sqrt{\kappa} < \rho < 0$, which are glued at $\rho = 0$. Metric (13.2) represents a sphere in cylindrical coordinates.

- (b) (i) Light rays move on curves (geodesics) satisfying,

$$c^2 = \left(\frac{dr}{dt} \right)^2 + r(t)^2 \left(\frac{d\varphi}{dt} \right)^2. \quad (13.8)$$

A simple case is, $d\varphi = 0$ (constant angle). In this case it is easy to solve (13.8),

$$r(t) = r(0) \pm ct \quad (\text{for } \varphi = \text{const.}) \quad (13.9)$$

A more general solution can be obtained by noticing that (13.1) is a flat space metric in cylindrical coordinates, for which geodesics in Cartesian coordinates are,

$$x(t) = x(0) \pm ct, \quad y(t) = y(0) \pm ct, \quad z(t) = z(0) \pm ct. \quad (13.10)$$

Next, making use of the coordinate transformations,

$$\varphi = \text{Arctan} \left(\frac{y}{x} \right), \quad r = \sqrt{x^2 + y^2} \quad (13.11)$$

the solutions (13.10) can be rewritten as,

$$\begin{aligned} \varphi(t) &= \text{Arctan} \left(\frac{r(0) \sin(\varphi(0)) \pm ct}{r(0) \cos(\varphi(0)) \pm ct} \right), \\ r(t) &= \sqrt{[r(0) \cos(\varphi(0)) \pm ct]^2 + [r(0) \sin(\varphi(0)) \pm ct]^2} \end{aligned} \quad (13.12)$$

The solution (13.9) is obtained by setting $\varphi(0) = 0$ in (13.12). The more general solutions with $\varphi = \text{const.}$ in (13.12) are obtained by rotating (13.9) by the angle $\varphi(0)$.

(b) (ii) Here we shall not construct the general solution. Instead, we shall use the hint that states that one can consider a solution in $\varphi = \text{const.}$, and other solutions can be obtained by making use of the rotational symmetry of the problem, *i.e.* by rotating the solution constructed for $\varphi = \text{const.}$ The light ray equation then becomes,

$$cdt = \pm \frac{d\rho}{\sqrt{1 - \kappa\rho^2}}. \quad (13.13)$$

It is useful to separate the $\kappa > 0$ and $\kappa < 0$ cases,

$$\begin{aligned} \rho(t) &= \rho(0) \pm \frac{1}{\sqrt{\kappa}} \sin [\sqrt{\kappa} ct], \quad (\kappa > 0) \\ \rho(t) &= \rho(0) \pm \frac{1}{\sqrt{-\kappa}} \sinh [\sqrt{-\kappa} ct], \quad (\kappa < 0). \end{aligned} \quad (13.14)$$

More general solutions can be obtained by rotating this solution by an angle φ_0 or in a different way. There are in fact three independent rotations one can perform on this geometry, but it is too early to try to explain that here.

(b) (iii) The geodesic equation for light rays can be written as,

$$c^2 = R^2 \left[\left(\frac{d\vartheta}{dt} \right)^2 + \sin^2(\vartheta) \left(\frac{d\varphi}{dt} \right)^2 \right]. \quad (13.15)$$

A simple solution is obtained when $\varphi = \text{const.}$, in which case,

$$\vartheta(t) = \vartheta(0) \pm \frac{ct}{R}, \quad (\varphi = \text{const.}). \quad (13.16)$$

This solution corresponds to geodesics originating in the north pole ($\vartheta(0) = 0$ and with the sign $+$ in (13.16)) and crossing each other at the south pole (at $t = \pi R/(c)$). More general solutions can be obtained by rotating (13.16). The 2 basic rotations are obtained by constructing rotation matrices in the (xy) , (yz) or (zx) -plane, and a general rotation is a composition of these three rotations.

(c) (i) In this case the manifold is, $\mathcal{M} = \mathbf{R} \times \mathbf{R}^2$, and a disc D_R of radius R has as the future domain of dependence $D^+(D_R)$ the region bounded by D_R in the past ($t = 0$) and the cone $r(t) = R - ct, 0 \leq t \leq R/c, \varphi \in [0, 2\pi)$. Similarly, the past domain of dependence is the region bounded by D_R in the future ($t = 0$) and the cone $r(t) = R + ct, 0 \geq t \geq -R/c, \varphi \in [0, 2\pi)$.

(c) (ii) When $\kappa > 0$ the manifold is, $\mathcal{M} = \mathbf{R} \times S^2$, with the sphere S^2 of radius, $R = 1/\sqrt{\kappa}$. The region is the polar cap of radius $\rho_0 = R \sin(\vartheta_0)$, where $R = 1/\sqrt{\kappa}$ and ϑ_0 azimuthal angle. This region can be also characterized by an arc of length

$$\ell_0 = R\vartheta_0 = \frac{\arcsin(\sqrt{\kappa}\rho_0)}{\sqrt{\kappa}}. \quad (13.17)$$

The future domain of dependence D^+ of the polar cap is then the region bounded in the past ($t = 0$) by the polar cap $\rho \leq \rho_0$ in the past ($t = 0$) and by $\ell(t) = \ell_0 - ct$ ($0 \leq t \leq \ell_0/c$), $\varphi \in [0, 2\pi)$. The past domain of dependence D^- is then the region bounded in the future ($t = 0$) by the polar cap $\rho \leq \rho_0$, $0 \leq \varphi, 2\pi$ in the future ($t = 0$) by $\ell(t) = \ell_0 + ct$ ($0 \geq t \geq -\ell_0/c$), $\varphi \in [0, 2\pi)$.

When $\kappa < 0$ the manifold is, $\mathcal{M} = \mathbf{R} \times \mathbf{H}^2$, with the throat radius, $R = 1/\sqrt{-\kappa}$. In this case it is useful to transform into the coordinates, $\sinh(\psi) = \sqrt{-\kappa}\rho$, in terms of which the metric (ii) becomes,

$$ds^2 = -c^2 dt^2 + \frac{1}{-\kappa} [d\psi^2 + \sinh^2(\psi) d\varphi^2]. \quad (13.18)$$

When viewed in this metric, $\rho = \frac{1}{\sqrt{-\kappa}} \sinh(\psi)$, or $\psi = \operatorname{arcsinh}(\sqrt{-\kappa}\rho) = \ln \left[\sqrt{-\kappa}\rho + \sqrt{-\kappa\rho^2 + 1} \right]$, $\psi_0 = \operatorname{arcsinh}(\sqrt{-\kappa}\rho_0)$. The future and past lightcones are given by $ds^2 = 0$, which for $\varphi = \varphi_0$ become, $\psi(t) = \psi_0 \pm ct$. With this in mind, the future domain of dependence D^+ of the set $0 \leq \psi \leq \psi_0, 0 \leq \varphi \leq 2\pi$ is given by the region whose boundaries are in the past (at $t = 0$) that set and $\psi(t) = \psi_0 - ct$ ($0 \leq t \leq \psi_0/c$), $\varphi \in [0, 2\pi)$. The past domain of dependence D^- is given by the region whose future boundary (at $t = 0$) is that set and $\psi(t) = \psi_0 + ct$ ($0 \geq t \geq -\psi_0/c$), $\varphi \in [0, 2\pi)$.

(c) (iii) Finally, by observing that these coordinate represent the sphere considered in part (ii), with the identifications, $R = 1/\sqrt{\kappa}$, $\sin(\vartheta_0) = \sqrt{\kappa}\rho_0$, the future and past domains of dependence can be read off from part (ii).

(d) (i) Examples of Cauchy surfaces for part (i) are the planes, $\{(x, y) | x, y \in \mathbf{R}\} \equiv \{(r, \varphi) | r > 0, \varphi \in [0, 2\pi)\}$ for any $t = t_0$ ($t_0 \in \mathbf{R}$).

Building on part (c) (ii) we see that Cauchy surfaces are the whole spheres S^2 (for $\kappa > 0$) and hyperboloids $\mathbf{H}^2$ (for $\kappa < 0$) at $t = t_0$ with $t_0 \in \mathbf{R}$, respectively.

For the metric (iii) again Cauchy surfaces are the whole spheres S^2 at any $t = t_0$ ($t_0 \in \mathbf{R}$).

We leave as an exercise to the students to sketch all of the surfaces and regions discussed in parts (c) and (d) of this problem.

■ PROBLEM 14 Integration measure in terms of the ϵ -tensor

Recall from last week that the ϵ -tensor is given by

$$\epsilon \equiv \sqrt{|g|} \frac{1}{n!} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge dx^{\mu_2} \wedge \dots \wedge dx^{\mu_n} \equiv \sqrt{|g|} d^n x \quad (14.1)$$

The volume of an n -manifold M is defined as the integral

$$V[M] \equiv \int_M \epsilon. \quad (14.2)$$

For a scalar field $\phi : M \rightarrow \mathbb{R}$, the integral over M is then defined as

$$I_M[\phi] \equiv \int_M \epsilon \phi. \quad (14.3)$$

(a) Consider the unit 2-sphere S^2 . Calculate ϵ and the 'volume' $V[S^2]$.

Furthermore, given $\phi, \psi : S^2 \rightarrow \mathbb{R}$, $\phi(x, y, z) = x$ and $\psi(x, y, z) = x^2$, calculate the integrals $I_{S^2}[\phi]$ and $I_{S^2}[\psi]$.

(b) Now consider the torus T^2 , with radii r and R ($r < R$). Calculate $V[T^2]$ and the integrals $I_{T^2}[\phi]$ and $I_{T^2}[\psi]$.

(c) Find the area of a cone, with deficit angle $2\pi\Delta$ and length L .

Hint: Show first that the metric can be written as, $ds^2 = d\rho^2 + (1 - \Delta)^2 \rho^2 d\phi^2$, ($0 \leq \rho \leq L$, $0 \leq \phi < 2\pi$), where $0 \leq \Delta < 1$ is a constant. What is the physical meaning of Δ ?

SOLUTION: (a) Working in spherical coordinates (see (13.3) with $R = 1$), the epsilon tensor for a unit sphere is,

$$\epsilon = \sin(\vartheta) \frac{1}{2} [d\vartheta \wedge d\varphi - d\varphi \wedge d\vartheta] = \sin(\vartheta) d\vartheta \wedge d\varphi, \quad (0 \leq \vartheta \leq \pi, 0 \leq \varphi < 2\pi). \quad (14.4)$$

The corresponding volume is,

$$V(S^2) = \int_{S^2} \epsilon = \int_0^\pi \sin(\vartheta) d\vartheta \wedge \int_0^{2\pi} d\varphi = 4\pi \quad (14.5)$$

Note that, had one reversed the order of integration, one would have obtained -4π . This means that the volume defined by means of the Levi-Civita tensor is oriented, *i.e.* a sign plus or minus is associated with it.

Next, integrating $\phi = x = \cos(\vartheta) \cos(\varphi)$ over S^2 gives,

$$I_{S^2}(\phi) = \int_{S^2} \phi \epsilon = \int_0^\pi \sin(\vartheta) \cos(\vartheta) d\vartheta \wedge \int_0^{2\pi} \cos(\varphi) d\varphi = 0 \quad (14.6)$$

which follows from, $\int_0^{2\pi} \cos(\varphi) d\varphi = 0$.

Finally, integrating $\psi = x^2 = \cos^2(\vartheta) \cos^2(\varphi)$ over S^2 gives,

$$I_{S^2}(\psi) = \int_{S^2} \psi \epsilon = \int_0^\pi \sin(\vartheta) \cos^2(\vartheta) d\vartheta \wedge \int_0^{2\pi} \cos^2(\varphi) d\varphi = \frac{2\pi}{3}, \quad (14.7)$$

which follows from, $\int_0^{2\pi} \cos^2(\varphi) d\varphi = \int_0^{2\pi} [\frac{1}{2} + \frac{1}{2} \cos(2\varphi)] d\varphi = \pi$ and $\int_0^\pi \sin(\vartheta) \cos^2(\vartheta) d\vartheta = \int_{-1}^1 \cos^2(\vartheta) d \cos(\vartheta) = 2/3$.

(b) The embedding coordinates on a 2-torus T^2 are,

$$\begin{aligned} x &= (R + r \cos(\vartheta)) \cos(\varphi), & (0 \leq \varphi < 2\pi) \\ y &= (R + r \cos(\vartheta)) \sin(\varphi), & (0 \leq \vartheta < 2\pi) \\ z &= r \sin(\vartheta), \end{aligned} \quad (14.8)$$

which imply the line element,

$$ds^2 = r^2 d\vartheta \otimes d\vartheta + (R + r \cos(\vartheta))^2 d\varphi \otimes d\varphi \quad (14.9)$$

The Levi-Civita tensor is,

$$\epsilon = r(R + r \cos(\vartheta)) d\vartheta \wedge d\varphi, \quad (14.10)$$

from which we can calculate the volume of T^2 ,

$$V(T^2) = \int_{T^2} \epsilon = r \int_0^{2\pi} (R + r \cos(\vartheta)) d\vartheta \wedge \int_0^{2\pi} d\varphi = 4\pi^2 r R \quad (14.11)$$

Next, integrating $\phi = x = (R + r \cos(\vartheta)) \cos(\varphi)$ over T^2 gives,

$$I_{T^2}(\phi) = \int_{T^2} \phi \epsilon = r \int_0^{2\pi} (R + r \cos(\vartheta))^2 d\vartheta \wedge \int_0^{2\pi} \cos(\varphi) d\varphi = 0 \quad (14.12)$$

which follows from, $\int_0^{2\pi} \cos(\varphi) d\varphi = 0$.

Finally, integrating $\psi = x^2 = (R + r \cos(\vartheta))^2 \cos^2(\varphi)$ over T^2 gives,

$$I_{T^2}(\psi) = \int_{T^2} \psi \epsilon = r \int_0^{2\pi} (R + r \cos(\vartheta))^3 d\vartheta \wedge \int_0^{2\pi} \cos^2(\varphi) d\varphi = \pi^2 (2R^2 + 3r^2) r R, \quad (14.13)$$

which follows from, $\int_0^{2\pi} \cos^2(\varphi) d\varphi = \pi$ and $\int_0^{2\pi} (R + r \cos(\vartheta))^3 d\vartheta = \int_0^{2\pi} (R^3 + 3Rr^2 \cos^2(\vartheta)) d\vartheta = 2\pi R^3 + 3\pi Rr^2$.

(c) The flat coordinates on the cone $\mathbf{C}_L$ can be described by the line element,

$$ds^2 = d\rho^2 + \rho^2 d\Phi^2, \quad (0 \leq \rho \leq L, 0 \leq \Phi < 2\pi(1 - \Delta)). \quad (14.14)$$

These coordinates are the cylindrical coordinates of a plane, obtained by cutting a wedge of the cone and flattening it onto the plane. Now, performing the coordinate transformation, $\phi = (1 - \Delta) \Phi$ yields,

$$ds^2 = d\rho^2 + (1 - \Delta)^2 \rho^2 d\phi^2, \quad (0 \leq \rho \leq L, 0 \leq \phi < 2\pi). \quad (14.15)$$

It is instructive to find the volume of the cone in both coordinates, and convince oneself that the volume is independent on the coordinates. This can be thanked to the tensorial nature of the Levi-Civita tensor. Firstly, using the coordinates (14.14) one obtains,

$$V(C_L) = \int_{C_L} \epsilon = \int_0^L \rho d\rho \wedge \int_0^{2\pi(1-\Delta)} d\phi = 2\pi(1 - \Delta)L. \quad (14.16)$$

On the other hand, using the coordinates (14.15) yields,

$$V(C_L) = \int_{C_L} \epsilon = \int_0^L (1 - \Delta) \rho d\rho \wedge \int_0^{2\pi} d\phi = 2\pi(1 - \Delta)L. \quad (14.17)$$

which is identical to (14.16), as it should be. Regarding the range of Δ , $\Delta < 1$ because at the point $\Delta = 1$ the manifold reduces to a half-line, $\infty > \rho \geq 0$, *i.e.* the dimensionality of the manifold changes from 2 to 1. Therefore, the manifold $\Delta = 1$ is not smoothly related to $\Delta < 1$. In principle, $\Delta < 0$ (corresponding to cones with a surplus angle), and there is no limit to how large that surplus can be. For simplicity, in this problem, we choose $\Delta \geq 0$. From gravity in two spatial dimensions, we know that point masses m are objects that induce an angle deficit Δ on the manifold, such that $\Delta \propto m$. On physical grounds we know that masses can be positive, but not negative, which then excludes $\Delta < 0$.

■ **PROBLEM 15** Christoffel symbols and covariant derivatives I.

In the lecture it was shown that Christoffel symbols compatible with the metric have the following form

$$\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}g^{\alpha\beta}\left(\partial_{\mu}g_{\nu\beta} + \partial_{\nu}g_{\mu\beta} - \partial_{\beta}g_{\mu\nu}\right). \quad (15.1)$$

- (a) Show that Christoffel symbols are not tensors, *i.e.* show that under coordinate transformation $x \rightarrow \tilde{x}$ they transform in the following manner

$$\tilde{\Gamma}_{\mu\nu}^{\alpha} = \frac{\partial \tilde{x}^{\alpha}}{\partial x^{\beta}} \frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} \Gamma_{\rho\sigma}^{\beta} + \frac{\partial \tilde{x}^{\alpha}}{\partial x^{\beta}} \frac{\partial^2 x^{\beta}}{\partial \tilde{x}^{\mu} \partial \tilde{x}^{\nu}}. \quad (15.2)$$

(Note that this is the reason why we are not careful about index placement with Christoffel symbols). Are they tensors under linear coordinate transformations?

- (b) Using the result of problem (a) and the definition of the covariant derivative, show explicitly that the following two quantities transform as tensors

$$(i) \quad \nabla_{\mu}\omega_{\nu}, \quad (ii) \quad \nabla_{\mu}V^{\nu}, \quad (15.3)$$

and that $\nabla_{\mu}V^{\mu}$ is a scalar quantity.

SOLUTION: (a) Christoffel symbols are tensors under linear coordinate transformations because the non-tensorial part of Eq. (15.2) vanishes for linear coordinate transformations.

To prove (15.2), recall first that the metric and inverse metric transform as,

$$\tilde{g}_{\mu\nu}(\tilde{x}) = \frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} g_{\rho\sigma}(x), \quad \tilde{g}^{\alpha\beta}(\tilde{x}) = \frac{\partial \tilde{x}^{\alpha}}{\partial x^{\eta}} \frac{\partial \tilde{x}^{\beta}}{\partial x^{\theta}} g^{\eta\theta}(x). \quad (15.4)$$

Inserting these into the definition of $\tilde{\Gamma}_{\mu\nu}^{\alpha}$ (*cf.* Eq. (15.1)) one obtains,

$$\tilde{\Gamma}_{\mu\nu}^{\alpha}(\tilde{x}) = \frac{1}{2} \frac{\partial \tilde{x}^{\alpha}}{\partial x^{\eta}} \frac{\partial \tilde{x}^{\beta}}{\partial x^{\theta}} g^{\eta\theta}(x) \left[\tilde{\partial}_{\mu} \left(\frac{\partial x^{\rho}}{\partial \tilde{x}^{\nu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\beta}} g_{\rho\sigma}(x) \right) + \tilde{\partial}_{\nu} \left(\frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\beta}} g_{\rho\sigma}(x) \right) - \tilde{\partial}_{\beta} \left(\frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} g_{\rho\sigma}(x) \right) \right], \quad (15.5)$$

where $\tilde{\partial}_{\mu} = \frac{\partial x^{\gamma}}{\partial \tilde{x}^{\mu}} \partial_{\gamma}$, and analogously for $\tilde{\partial}_{\nu}$ and $\tilde{\partial}_{\beta}$. Acting the derivatives inside the square brackets gives two types of contributions, one can be identified as the tensorial part of the Christoffel symbol transformation in (15.2), while the other is the non-tensorial part of the transformation.

- (b) (i) From,

$$\nabla_{\mu}\omega_{\nu} = \partial_{\mu}\omega_{\nu} - \Gamma_{\mu\nu}^{\alpha}\omega_{\alpha} \rightarrow \tilde{\nabla}_{\mu}\tilde{\omega}_{\nu} = \tilde{\partial}_{\mu}\tilde{\omega}_{\nu} - \tilde{\Gamma}_{\mu\nu}^{\alpha}\tilde{\omega}_{\alpha}, \quad (15.6)$$

we can write,

$$\tilde{\nabla}_{\mu}\tilde{\omega}_{\nu} = \frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \partial_{\rho} \left[\frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} \omega_{\sigma} \right] - \left[\frac{\partial \tilde{x}^{\alpha}}{\partial x^{\beta}} \frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} \Gamma_{\rho\sigma}^{\beta} + \frac{\partial \tilde{x}^{\alpha}}{\partial x^{\beta}} \frac{\partial^2 x^{\beta}}{\partial \tilde{x}^{\mu} \partial \tilde{x}^{\nu}} \right] \left[\frac{\partial x^{\gamma}}{\partial \tilde{x}^{\alpha}} \omega_{\gamma} \right], \quad (15.7)$$

Acting the derivative in the first term and multiplying the other terms gives,

$$\tilde{\nabla}_{\mu}\tilde{\omega}_{\nu} = \left[\frac{\partial^2 x^{\sigma}}{\partial \tilde{x}^{\mu} \partial \tilde{x}^{\nu}} \omega_{\sigma} + \frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} \partial_{\rho} \omega_{\sigma} \right] - \left[\frac{\partial x^{\rho}}{\partial \tilde{x}^{\mu}} \frac{\partial x^{\sigma}}{\partial \tilde{x}^{\nu}} \Gamma_{\rho\sigma}^{\beta} \omega_{\beta} - \frac{\partial^2 x^{\beta}}{\partial \tilde{x}^{\mu} \partial \tilde{x}^{\nu}} \omega_{\beta} \right]. \quad (15.8)$$

It is now easy to see that the non-tensorial parts in (15.8) cancel, and the remaining pieces combine into a covariant derivative, yielding,

$$\tilde{\nabla}_\mu \tilde{\omega}_\nu = \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} \nabla_\rho \omega_\sigma. \quad (15.9)$$

(b) (ii) From,

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\alpha}^\nu V^\alpha \rightarrow \tilde{\nabla}_\mu \tilde{V}^\nu = \tilde{\partial}_\mu \tilde{V}^\nu + \tilde{\Gamma}_{\mu\alpha}^\nu \tilde{V}^\alpha, \quad (15.10)$$

we can write,

$$\tilde{\nabla}_\mu \tilde{V}^\nu = \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \partial_\rho \left[\frac{\partial \tilde{x}^\nu}{\partial x^\sigma} V^\sigma \right] + \left[\frac{\partial \tilde{x}^\nu}{\partial x^\beta} \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial x^\sigma}{\partial \tilde{x}^\alpha} \Gamma_{\rho\sigma}^\beta + \frac{\partial \tilde{x}^\nu}{\partial x^\beta} \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\alpha} \right] \left[\frac{\partial \tilde{x}^\alpha}{\partial x^\gamma} V^\gamma \right], \quad (15.11)$$

Acting the derivative in the first term and multiplying the other terms gives,

$$\tilde{\nabla}_\mu \tilde{V}^\nu = \left[\frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial^2 \tilde{x}^\nu}{\partial x^\rho \partial x^\sigma} V^\sigma + \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} \partial_\rho V^\sigma \right] + \left[\frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} \Gamma_{\rho\gamma}^\sigma V^\gamma + \frac{\partial \tilde{x}^\nu}{\partial x^\beta} \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\alpha} \frac{\partial \tilde{x}^\alpha}{\partial x^\gamma} V^\gamma \right]. \quad (15.12)$$

It is now easy to see that the second and third term combine into the desired relation,

$$\tilde{\nabla}_\mu \tilde{V}^\nu = \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} \nabla_\rho V^\sigma. \quad (15.13)$$

The non-tensorial parts in (15.12)

$$\left[\frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial^2 \tilde{x}^\nu}{\partial x^\rho \partial x^\sigma} + \frac{\partial \tilde{x}^\nu}{\partial x^\beta} \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\alpha} \frac{\partial \tilde{x}^\alpha}{\partial x^\sigma} \right] V^\sigma. \quad (15.14)$$

ought to cancel for arbitrary V^σ . To see that, note that the second term in (15.14) can be written as,

$$\begin{aligned} \frac{\partial \tilde{x}^\nu}{\partial x^\beta} \frac{\partial}{\partial x^\sigma} \left[\frac{\partial x^\beta}{\partial \tilde{x}^\mu} \right] &= \frac{\partial}{\partial x^\sigma} \left[\frac{\partial \tilde{x}^\nu}{\partial x^\beta} \frac{\partial x^\beta}{\partial \tilde{x}^\mu} \right] - \frac{\partial}{\partial x^\sigma} \left[\frac{\partial \tilde{x}^\nu}{\partial x^\beta} \right] \frac{\partial x^\beta}{\partial \tilde{x}^\mu} = \frac{\partial}{\partial x^\sigma} [\delta_\mu^\nu] - \frac{\partial}{\partial x^\sigma} \left[\frac{\partial \tilde{x}^\nu}{\partial x^\rho} \right] \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \\ &= - \frac{\partial^2 \tilde{x}^\nu}{\partial x^\rho \partial x^\sigma} \frac{\partial x^\rho}{\partial \tilde{x}^\mu}, \end{aligned} \quad (15.15)$$

which is precisely what we need to cancel the first term in (15.12), completing the proof.

Finally, to show that $\nabla_\mu V^\mu$ transforms as a scalar, multiply Eq. (15.13) by δ_ν^μ to obtain,

$$\tilde{\nabla}_\mu \tilde{V}^\mu = \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial \tilde{x}^\mu}{\partial x^\sigma} \nabla_\rho V^\sigma = \delta_\sigma^\rho \nabla_\rho V^\sigma = \nabla_\rho V^\rho, \quad (15.16)$$

thus $\nabla_\mu V^\mu$ indeed transforms as a scalar.